

Javad J.C. Aman

Data Scientist • Applied AI/ML, GenAI, Forecasting & Anomaly Detection
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SUMMARY

Data Scientist with senior-level impact and 6+ years in data science and machine learning, delivering production AI/ML across two lines of business, finance and manufacturing, with multimillion-dollar value measured through forecasting and anomaly-detection systems. Practical experience designing LLM solutions, including RAG, agent workflows, and LLM classification, deployed for internal use, alongside supervised and unsupervised machine learning and causal inference. Owns the full lifecycle from ideation to production, applying strong software engineering fundamentals, including version control, CI/CD, and MLOps, while aligning technical execution with business strategy through client and stakeholder relationships.

WORK EXPERIENCE

Caterpillar Inc.

Data Scientist II

Dallas, TX

Apr 2023 – Present

- Built **anomaly-detection and fact-checking systems** for finance and manufacturing, applying **NLP to unstructured data** and combining statistical methods with **LLMs (RAG, agent workflows, vector search)**, deployed for internal use, to flag irregularities in journal entries and financial statements, **eliminating hours of manual analyst review**.
- Built and evaluated a production **RAG pipeline** (Azure AI Search, FAISS) with a custom **evaluation harness** for **performance tracking and value measurement**, including a golden dataset, **RAGAS faithfulness and context-precision scoring**, and a human-calibrated LLM judge, surfacing a grounding failure mode and improving output faithfulness.
- Drove **multimillion-dollar gains** in forecast value by developing **high-volume, near-real-time revenue and cost-optimization forecasting** across business sectors, segments, and regions, combining **CatBoost and GPBoost ensembles** with **causal-effect analysis** and **A/B testing**.
- Enabled reliable, governed model deployment by owning **end-to-end data pipelines** across **Snowflake, Azure, and AWS**, writing modular, production-grade Python with **version control (Git)**, and building **MLOps** practices, including **model monitoring, drift detection, and CI/CD**, from prototype to production.
- Replaced slow, inconsistent manual pricing-code labeling with a **reproducible, incremental pipeline** by designing a **14-category LLM classifier** (zero-shot / few-shot) and migrating it from local prototype to **enterprise Azure AI Foundry**.
- Identified the key drivers behind sales and cost outcomes through **feature-importance and interpretability analysis**, giving finance stakeholders decision-support dashboards showing exactly what was moving their forecasts.
- Mentored **interns and junior team members** through code review and hands-on pairing on model design and deployment, leading collaborative projects with them from scoping through delivery.

Southern Methodist University

Research Assistant

Dallas, TX

Jan 2019 – Mar 2023

- Modeled **post-COVID mobility shifts** by processing **multi-terabyte GPS data** on SMU's **ManeFrame HPC cluster**, applying **gradient-boosting and clustering** to identify significant behavior changes across Dallas-area residents.
- Applied **NLP and CNNs (TensorFlow/Keras)** to **100,000+** tweets and app reviews on micromobility services, translating rider-sentiment findings into recommendations adopted by city-planning partners.

High Street Consulting

Mobility Data Analyst

Pittsburgh, PA

Jan 2022 – May 2022

- Ensured client satisfaction by collaborating with a **six-engineer team** to scope project requirements, needs, and deadlines throughout delivery.
- Developed and validated a **Random Forest model** for **traffic-speed forecasting** and built interactive **Tableau and Power BI dashboards** on **35M+** row federal travel-monitoring datasets, reducing manual data-collection requirements and revealing spatio-temporal traffic patterns.

EDUCATION

Ph.D., **Transportation Engineering**, Southern Methodist University, Dallas, TX

2023

Dissertation: Machine learning techniques to reveal the mobility patterns of residents using cellphone location data.

M.S., **Transportation Engineering**, Sharif University of Technology, Tehran, Iran

2015

ADDITIONAL TRAINING

Applied Machine Learning Program, Columbia University, New York, NY

2022

SELECTED PUBLICATIONS

Aman, J.J.C., & Smith-Colin, J. (2022). Application of Crowdsourced Data to Infer User Satisfaction with Mobility as a Service (MaaS). *Journal of Transportation Research Interdisciplinary Perspectives*.

Aman, J.J.C., Smith-Colin, J., & Zhang, W. (2021). Listen to E-Scooter Riders: Mining Factors of Rider Satisfaction from App Reviews. *Transportation Research Part D*.

SKILLS

- GenAI & NLP:** RAG, agent workflows, vector search (FAISS, Azure AI Search), agentic & multi-agent systems (LangChain, LangGraph), OpenAI SDK, LLM evaluation & benchmarking, fine-tuning, Hugging Face, knowledge graphs, NLP, Azure AI Foundry
- ML & Statistics:** Supervised & unsupervised machine learning, CatBoost, GPBoost, forecasting, anomaly detection, causal inference, A/B testing, deep learning (TensorFlow, PyTorch)
- Cloud & Data Engineering:** Snowflake, Azure, AWS (SageMaker, Lambda, Glue, Step Functions), Docker, Kubernetes, CI/CD, MLOps, version control (Git), HPC
- Programming:** Python, R, SQL
- Visualization:** Power BI, Tableau